
Suggested Teaching Guidelines for
Advanced Programming for AI (Java Programming)
PGCP-AI August 2026

Duration: 90 Hours: 36 Theory (T) + 30 Lab (L) + 24 Self-Learning (SL)

Objective: To introduce the student to Java programming

Prerequisites: Knowledge of basic Programming Concepts and OOP

Evaluation method: Theory exam – 40%, Lab exam – 40%, Internal Assessment– 20%

List of Course Material

Text Book: No specific courseware for the module, faculty may share some coursematerials.

Reference Book:

1. Object-Oriented Programming and Data Structures using Java, 2026 edition, Cay S. Horstmann / Wiley
2. Java The Complete Reference, 9th edition, Herbert Schildt / McGraw Hill
3. Murach's Java Programming, 4th edition, Joel Murach

Note:

- ***Lab assignments are indicative; faculty shall provide adequate assignments for better practice.***
- ***Faculties are advised to give at least one lab assignment as end-to-end application.***
- ***This module adopts Java 17 (LTS) as the programming standard for all sessions, lab work and evaluations.***

Session 1 & 2: (4T+ 2L+ 2SL)

Theory

- Introduction to JVM Architecture
- Java Basics: Overview of Java, Features of Java, Scope of variables
- Object Oriented Concepts
- JDK and its usage (Java Compiler, Java Runtime, Java Debugger, Java doc)
- Working with Data Types: Structure of a Java Class, Importing Packages
- Wrapper classes (Boolean, Double and Integer)
- Operators (Unary, Binary, Arithmetic, Assignment, Compound, Relational, Logical, Equality) and Control Statements (if, if-else, for, while, switch, do-while, break and continue)

Sessions 3 & 4: (6T + 6L + 4SL)

Theory

- Packages
- Arrays
- Understanding of String Class, String Builder Class
- Methods and Encapsulation: Methods, Access Modifiers, Method Overloading, Passing Data, Creating Constructors, Immutable Classes
- Class Inheritance, Abstract Classes, Interface and Implementation classes.
- Understanding Polymorphism: Object vs Reference, Object Casting, Virtual Methods, Method Overriding

*Suggested Teaching Guidelines for***Advanced Programming for AI (Java Programming)****PGCP-AI August 2026****Session 5: (4T +4L +2SL)****Theory**

- Exception-Handling: Basics, Role of Exceptions, Types
- Using try and catch, Multiple Catch, Nested try (throw, throws, finally)
- Built-in Exceptions, Runtime Exceptions Checked Exceptions, Errors
- Creating own Exception Subclasses

Session 6: (2T +2L +2SL)**Theory**

- Enumerations, Auto boxing
- Java API: java.util, java.lang, java.math
- Generics and Collections - Overview

Sessions 7 & 8 (4T + 4L +4SL)**Theory**

- Functional Programming Overview
- Functional Interfaces
- Explore java.util.function package : Predicate, Map, Consumer, Supplier
- Lambda Expressions
- Impact of Functional programming upon Collection Framework

Lab Assignment:

- Create an appropriate data structures to store your employee object and use the java.util.package properties.

Sessions 9 & 10 (6T + 6L + 4SL)**Theory**

- Introduction to Streams
- Streams vs. Collections
- java.util.stream.Stream API
- Types of Primitive Streams : IntStream,LongStream,DoubleStream & its API
- Different operations on streams : filter, map, reduce, sort, flatMap, anyMatch, count, boxing.
- Overview of Java Date Time API

Sessions 11 & 12: (4T + 4L +2SL)**Theory**

- Java Concurrency: Using threads in Java, Life cycle of thread
- Advantages and issues
- Thread class, thread groups
- The Runnable interface
- Synchronization

Session 13: (2T +2L)**Theory: Reflection in Java**

- Why Java Reflection
- Basic Reflection API for finding out details of the class name, super classes & interfaces.

Suggested Teaching Guidelines for

Advanced Programming for AI (Java Programming)

PGCP-AI August 2026

Session 14: (2 T + 2SL)

Theory: Introduction to Node

- Node.js
- Browser JS vs. Node.js
- Node.js REPL

Understand how reflection enables frameworks like Spring and Hibernate

Learn to inspect class metadata, methods, and interfaces at runtime

Identify performance and security considerations of reflection

Session 15: (2 T + 2SL)

Theory: Introduction to Spring Frame Work

- About Spring Framework
- Overview of Spring Architecture
- Spring MVC architecture
- Spring Modules Overview